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RESEARCH INTERESTS	Economics of education, development economics, behavioral economics and machine learning	
ACADEMIC POSITIONS	<i>Postdoctoral Researcher</i> ifo Center for the Economics of Education	2026-
ACADEMIC BACKGROUND	<i>Ph.D., Economics</i> Massachusetts Institute of Technology, MIT - Advisors: Prof. Sendhil Mullainathan, Prof. Esther Duflo and Prof. Frank Schilbach <i>M.A. International Education Policy Analysis</i> Stanford University - Thesis: I Want You to Teach My Children: The Effect of Changing the Teacher Hiring Process on Parental School Choice in the Mexican State of Zacatecas <i>B.A. Economics</i> Instituto Tecnológico Autónomo de México, ITAM - Thesis: The Effect of Universal Scholarships on High School Dropout Rates. Evidence from Mexico - Graduated with highest honors	2020-2026 2019-2020 2015-2019
EXPERIENCE	<i>Co-Founder</i> Intendere SAPI - Digital tool that allows institutions to set up large-scale tutoring programs: with presence in Mexico and Peru - Co-founders: Fernanda Albo and Hans Ramírez <i>Co-Founder</i> Jovenes Ayudando a Niñas y Niños AC (Tutoring nonprofit) - Free online math tutoring for children in primary and middle schools in Mexico - Incorporated as a nonprofit organization in 2023 - Co-founders: Fernanda Albo, Bernardo García and Sebastián Guevara <i>Research assistant for Prof. Joshua Angrist and Prof. Parag Pathak</i> Blueprint labs <i>Research assistant for Prof. Eric Bettinger</i> Stanford Center for Education Policy Analysis (CEPA) <i>Research assistant for Prof. Enrique Seira</i> Centro de Investigación Económica (CIE)	2025 - present 2021 - present 2022 - 2023 2019 - 2020 2017 - 2019

GRANTS AND FELLOWSHIPS

Research grants

- *The Weiss Fund for Research in Development Economics* [2024](#)
- *J-WEL Grant in pK-12 Education Innovation* [2022](#)
- *The Weiss Fund for Research in Development Economics* [2022](#)
- *PPE Initiative Pilot Grant JPAL* [2021](#)
- *Schultz Fund at MIT*

Fellowships

- *Global Math Talent (NBER)* [2023-2024](#)
- *The Institute for Humane Studies Fellowship* [2023 and 2024](#)
- *Emma Krob Castle Graduate Fellow (MIT)* 2021-2022
- *Mexico Fellowship (MIT)* 2020-2021
- *Claudio X. Gonzalez Fellowship (Stanford University)* 2019-2020
- *CONACYT/FUNED Scholarship (Stanford University)* 2019-2020

Entrepreneurship

- *MIT Sandbox* [2022-2025](#)
- *MIT deltav* [2024](#)
- *Legatum Center for Development and Entrepreneurship (MIT)* [2023-2024](#)

TEACHING (at MIT)

- TA 14.03 Microeconomic Theory and Public Policy 2025
- TA 14.75 Political Economy and Economic Development 2023
- TA 14.01 Principles of Microeconomics 2022 and 2024

PUBLISHED PAPERS

- [Expanding Access to Tutoring: A Scalable Platform for Personalized Learning and Data-Driven Research](#) Joint work with Fernanda Albo, Bernardo Garcia and Sebastián Guevara *Proceedings of the 2025 AIED 26th International Conference on Artificial Intelligence in Education*
Expanding access to high-quality tutoring is critical for reducing educational disparities, yet scaling effective programs remains a challenge. We developed a platform that automates key logistical aspects of online tutoring, enabling large-scale implementation. Our system includes a real-time monitoring framework that tracks tutor activities. Prior research on online tutoring has shown positive effects on student learning. Building on this, we conducted a randomized controlled trial (RCT) in Mexico. We find that students assigned to tutoring improved their math scores by 0.14 standard deviations. Beyond tutoring delivery, the platform serves as a tool for research. Participating tutors upload class recordings. While this paper focuses on the tutoring intervention, we provide an overview of the platform's potential to facilitate large-scale RCTs. We also show some basic applications of machine learning tools to our data with the aim to analyze student-tutor interactions at scale, bridging the gap between quantitative and qualitative research in education.

- **TutorUp: What If Your Students Were Simulated? Training Tutors to Address Engagement Challenges in Online Learning** Joint work with Sitong Pan, Robin Schmucker, Bernardo García, Fernanda Albo, Hangxiao Zhu, Adam Teo, Meng Xia *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*

With the rise of online learning, many novice tutors lack experience engaging students remotely. We introduce TutorUp, a Large Language Model (LLM)-based system that enables novice tutors to practice engagement strategies with simulated students through scenario-based training. Based on a formative study involving two surveys ($N_1 = 86$, $N_2 = 102$) on student engagement challenges, we summarize scenarios that mimic real teaching situations. To enhance immersion and realism, we employ a prompting strategy that simulates dynamic online learning dialogues. TutorUp provides immediate and asynchronous feedback by referencing tutor-students online session dialogues and evidence-based teaching strategies from learning science literature. In a within-subject evaluation ($N = 16$), participants rated TutorUp significantly higher than a baseline system without simulation capabilities regarding effectiveness and usability. Our findings suggest that TutorUp provides novice tutors with more effective training to learn and apply teaching strategies to address online student engagement challenges.

WORKING PAPERS

- **Recording the impact of Social and Emotional Learning** Joint work with Fernanda Albo, Bernardo García

Social skills have been shown to predict earnings, teamwork, and productivity, and they are in increasing demand in the labor market. Schools play a central role in fostering these skills through their regular activities, but also increasingly, through social and emotional learning (SEL) interventions. Despite the widespread adoption of SEL programs, evidence on their effects on classroom dynamics is limited, as most studies rely on self-reports, survey measures, or classroom observation. We implement a randomized controlled trial (RCT) in an online setting where university students tutored elementary and middle school students in math. We randomized the delivery of a structured SEL curriculum of ten one-hour modules, compared to a control group receiving ten one-hour history modules. The SEL group reported improved rapport between tutors and students and increased math learning by 0.0848 standard deviations. To study its impact on classroom dynamics, we collected and transcribed more than 11,700 hours of tutoring sessions and applied a novel AI-based hypothesis generation method to the text. This analysis reveals that SEL-treated math classes featured a slower pace, less technical language, less frustration, more personal topics, and a less serious overall tone in the opening minutes of class. We benchmark these findings against prespecified outcomes from the prior SEL literature — engagement, classroom climate, and instructional quality — and show that the generated hypotheses capture additional variation beyond what these known measures explain. Together, these results demonstrate that SEL can significantly impact classroom behavior, relationships, and learning outcomes. They also showcase the potential of combining large-scale classroom recordings with hypothesis generation methods for better understanding of the dynamics of skill formation.

- **Can Single-Gender Classrooms Counteract Traditional Gender Beliefs? Evidence from a Tutoring RCT** Joint work with Bernardo García

This paper examines how the gender composition of students can moderate the impact of tutors' traditional gender beliefs on girls' math learning. We

conduct a randomized controlled trial in Mexico in which students are cross-randomized into single-gender or mixed-gender tutoring groups and assigned to tutors whose gender beliefs are measured through a survey. We find that girls in mixed-gender groups learn significantly less when assigned to a tutor with biased gender beliefs—by 0.10–0.14 standard deviations—whereas girls in single-gender groups seem to be less affected by tutors with biased gender beliefs. We rule out belief transmission as the main mechanism: post-intervention gender beliefs do not differ across treatment arms. Using an AI-based hypothesis generation approach applied to transcripts of tutoring sessions, we find that tutors with gender biased beliefs pose fewer questions to the class as a whole, and are more likely to address students individually and by name relative to non-TGB tutors—a pattern consistent with differential allocation of attention across students in mixed-gender settings. Conversely, when tutors do not hold traditional gender beliefs, girls in single-gender groups underperform relative to those in mixed-gender groups by 0.08–0.13 standard deviations. Taken together, these results suggest that the effectiveness of single-gender grouping as a policy intervention may depend on the gender beliefs of instructors.

- **Good Vibes in Class: A Tool to Detect Which Emotions Lead to More Learning** Joint work with Fernanda Albo, Bernardo García and Tobin South

We present a method to characterize the classroom environment through emotion detection from audio recordings. Using machine learning tools we build an emotion classifier using MFCC features of labeled voice clips and apply it to slices of more than 1,500 online class session records. We find that higher measurements of high-intensity emotions were significantly correlated with higher Teacher Value Added (TVA) estimates, determined using Math test scores of students before and after receiving tutoring. Secondly, we found that attendance metrics in the second class were highly correlated to the class environment in the first class. Finally, we found that higher-skilled tutors progressively increased high-intensity emotions as they had more sessions with their students.